

Zhinan (Mike) Liu

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Summary

MSEE candidate specializing in **embedded firmware, C/C++, and low-level optimization**. Experienced in building **performance-critical systems**, developing pipelines at scale, and applying **hardware/software integration skills** on microcontrollers, RTOS, and FPGA-based systems.

Education

University of Washington – MS in Electrical Engineering

Sep 2023 - Dec 2025 (Expected)

Relevant Coursework: Computer Architecture, Digital Systems Design with FPGAs, Data Structures & Algorithms, Embedded Software Design

University of Washington – BS in Biochemistry

Sep 2016 - June 2020

Experience

Researcher/Data Analyst, Harborview Medical Center– Seattle, WA

Feb 2021 - June 2024

- Engineered high-performance data pipelines (**Python, SQL, Jenkins**), automating dataset generation and saving **~20 staff hours a week**. Spearheaded creation of an automated **96,000+ patient dataset** for study of blood use in trauma.
- Applied statistical modeling and algorithm design (**Python, R**) to large-scale clinical datasets, improving predictive accuracy of geriatric readmission models (**F1 = 0.83**).
- Designed modular, testable code frameworks to ensure reproducibility and scalability across large datasets. Worked with Agile development and with multidisciplinary teams.
- Collaborated with cross-functional teams; contributed to **11 peer-reviewed publications**.

Research Assistant, Lieber Lab – Seattle, WA

August 2016 - August 2020

- Developed and optimized large-scale genomic analysis pipelines (**Python, Nextflow**), processing **400GB/week** efficiently.
- Built automation for CRISPR data analysis, demonstrating ability to handle high-throughput, performance-critical systems. Analyzed off-target CRISPR mutagenesis on **12,000+ genomic samples**.
- Documented pipeline design and maintained version control in Git, improving collaboration and long-term reproducibility.

Projects

esp_simd: High-Level SIMD Library for ESP32-S3

https://github.com/zliu43/esp_simd

High-level C/C++ library wrapping Xtensa SIMD for vector ops on ESP32-S3; designed for safe alignment/saturation and drop-in use with esp-idf.

- Implemented with hand-written, branchless ASM with zero-overhead loops
- Ops: Basic math ops including add/sub/mul/dotp across int8/16/32 & float32; benchmarks show ~5-10x speedups vs. scalar for int types and ~3-5x speedup for float types
- CMake examples and unit tests; ready for integration, including with esp-idf functions

Tech: C/C++, Xtensa ASM, esp-idf, CMake

rv3028c7_hal: Hardware Abstraction Layer for the MicroCrystal RV-3028 RTC

https://github.com/zliu43/rv3028c7_hal

- A lightweight C/C++ library providing a simple, portable API for interfacing with the MicroCrystal RV-3028 real-time clock (RTC)
- Features include: reading/writing time, setting registers, configuring alarms, formatting time, etc.

Skills

Programming Languages: C, C++, Python, Assembly (RISC-V, ARM, Xtensa), SystemVerilog, Bash

Embedded Systems & Firmware: ESP32, STM32, FreeRTOS, UART, I2C, SPI

Software Engineering & Tools: CMake, GCC, Git, Docker, Linux/Unix, Yocto, Microsoft Azure